AIR-GAPPED BITCOIN SIGNER · DOSSIER UNDER THE V1 CORE ARCHITECTURE

QuietKey QK2

The normative **Core Architecture v1**, the Decision Register, and eight working drafts — a calculator-disguised 2-of-3 signer whose recovery document is an ordinary encrypted webpage and whose bearer cards carry no PIN. The architecture is now selected and fixed; implementation and hardware validation remain stop-ship.

WALLET	SEEDS	CARDS	TRANSPORT	STATUS	DATE
P2WSH 2-of-3	24-word / 256-bit	B + C · bearer · no PIN	microSD + BBQr	selected · impl stop-ship	2026-08-18

SELECTED PRODUCT ARCHITECTURE; IMPLEMENTATION VALIDATION REMAINS STOP-SHIP

- QK2 is an interchangeable, air-gapped Bitcoin signer disguised as a retro calculator, storing **no wallet secret at rest**. The wallet is native **P2WSH 2-of-3** with independent Seeds A/B/C (24-word / 256-bit). Normal signing is **A1+B**; recovery is **A1+C or B+C**.
- **A1** is an ordinary printed webpage whose source-URL-looking footer contains a **ChaCha20-Poly1305** capsule of Seed A. **A2** (A1's decryption key) is generated at provisioning, copied to both bearer cards B and C, and wiped from the terminal. **B** holds signer B + A2 + descriptor D; **C** holds independent signer C + A2 + D. There is **no card PIN or terminal pairing** — cards are bearer objects; the 2-of-3 threshold is the control.
- **microSD (binary PSBT) and QR (BBQr) transport are both required** on the existing mechanics — **no mechanical redesign is authorized**. Descriptor disclosure is tiered: the Quantum Shelter kit is addresses-only, Simple/Inheritance may export the watch-only descriptor. QK2 is quantum-*prepared*, not quantum-proof.
- Card feasibility, the A1 capsule codec/capture, QR performance, the Rust signer implementation, recovery rehearsal, and QK2-HW-P0/RV3 evidence remain **blockers**. Contents are reference constructions at evidence level ≤ 2; nothing here is a validated implementation.

★ **THE AUTHORITY**

Start with the settled architecture

Everything below answers to one normative document. Where a working draft disagrees with the Core Architecture v1, the draft is being corrected, not the architecture.

QuietKey Core Architecture NORMATIVE V1

The settled design in twelve sections: one fixed 2-of-3 policy, the A1/A2/B/C custody model, the three-process privilege split, the ChaCha20-Poly1305 capsule, the BBQr/binary-SD PSBT profile, the Rust core hypothesis, and the five stop-ship gates.

Carries the 20-point reconciliation authority (§11) that the eight drafts and the register below are now aligned to.

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00 **WHAT QK2 IS**

A calculator-disguised 2-of-3 signer that hides in plain sight

CloakVault is defined by four inseparable ideas — **stealth, minimalism, strong encryption, hiding in plain sight**. The **terminal** stores no wallet secret at rest and is interchangeable. Seed A is restored transiently from **A1** (an ordinary encrypted webpage) using **A2** (its decryption key, duplicated on both bearer cards); the device co-signs a native **P2WSH 2-of-3** whose other legs are the smart cards **B** and **C**. Stealth is a first layer of defence — never a cryptographic factor and never a substitute for the 2-of-3 threshold. The drafts below reconcile the fixed hardware against the practices of the Ian Coleman BIP39 tool, SeedSigner, and ColdCard; the product architecture is selected, but feasibility is not claimed.

> **TRANSPORT & CAMERA**
>
> **microSD (binary PSBT v0) and QR (BBQr) are both required transports.** microSD is the lower-risk first implementation path; QR framing, capture, animated reassembly, screen export, and end-to-end performance remain a hardware/system validation blocker on the *existing* enclosure, optics, and 320×240 display — no mechanical redesign is authorized. The camera has two mode-locked uses only: **A1 footer capture** and **PSBT QR capture**; it must never auto-detect a seed QR or accept arbitrary QR classes.

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01 **THE NINE DOCUMENTS**

In reading order

REF reference computation **X-CHK** independent library cross-check **IMPL: none** production code not yet written/tested

HW: blocked target validation blocked

Core Architecture v1 NORMATIVE

The settled architecture and the reconciliation authority. Read this first; the register and drafts are subordinate to it. **One fixed policy; twelve sections; five gates; the 20-point punch-list.**

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00 Decision Register CONSISTENCY MATRIX

One row per decision, one column per document, each in exactly one state. The internal consistency matrix — subordinate to the Core Architecture v1 and to QK2-04. The regenerated conforming values and the retired constructions live here. **Where a document disagrees with the Core Architecture v1, the document is wrong.**

open →

01

QK2 Blueprint `START HERE`

The layered design, the custody topology, the three-process privilege split, best-practice extraction from the three reference codebases, and a threat matrix. **A malicious terminal in an A1+B session is one authorization trust domain — 2-of-3 is recovery resilience, not two-device authorization.**

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02

QKEC-1 Specification `X-CHK`

The provisioning entropy conditioner: 100-roll pure dice (Advanced) and a domain-separated HKDF-SHA256 multi-source mode (Recommended), with a stdlib verifier. 24 words / 256-bit only. **Cards contribute an RNG sample, not a secret; an account xprv (not a mnemonic) is provisioned — bench-proven, not inferred.**

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03

Standards Register `X-CHK`

Each BIP/SLIP/RFC with its *standards status*, QK2 adoption status, and test status — plus the `m/48'/0'/0'/2'` derivation as an adopted convention with interop tests pending. **Native P2WSH, sortedmulti, mainnet-only, Taproot out; ChaCha20-Poly1305, BBQr, exact RFC-6979. BIP-146 removed (withdrawn).**

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04

PSBT Checklist `X-CHK`

The ordered sign-or-reject pipeline for a **PSBT v0** profile — binary on microSD, BBQr on QR — checks classified fatal/warning, with a reference attack corpus (embit-built, not production-tested). **7 reference fixtures — 1 must-sign, 6 must-reject; no universal 10% fee rule; the negative corpus needs expanding.**

open →

05

Recovery Decision `POLICY SELECTED`

The selected recovery architecture — A1 ciphertext + A2 on bearer cards B and C, native 2-of-3, kit tiers (Simple / Inheritance /

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Quantum Shelter) — against a loss/rotation matrix. **The policy is selected; a real recovery rehearsal (Gate E) remains stop-ship.**

06

Card Protocol `HW: blocked`

The minimal bearer-card APDU surface — GET_A2, SIGN, and atomic provisioning of an account xprv + A2 + D — with no PIN and no custom secure channel in v1, gated on a real card capability matrix that does not yet exist. **No bespoke channel in v1; anti-exfil deferred; card feasibility (Gate B) is unproven.**

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07

Recovery Document `REF`

The printed footer-token coding: a **ChaCha20-Poly1305** capsule through Reed-Solomon to a **custom base32** token (not standard Bech32). Round-trip reference codec + erasure-first OCR. **AEAD = ChaCha20-Poly1305 with a fresh stored nonce; RS is never authenticity; human trials are open (Gate A).**

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08

Component Map `IMPL: none`

A provisional component map around the three-process boundary (qk-core / qk-io / qk-decoy), the Rust core + narrow libsecp FFI, and the A2 lifecycle — with the physical / software-privilege / logical-module distinction corrected. **Reference vectors ≠ production conformance; no module has passed target validation.**

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02 WHY THE VECTORS ARE CONSISTENT

One reference computation, cross-checked

The drafts share one arithmetic spine: the same seeds flow from entropy to a signature, and each stage was reproduced against an independently published constant or a second library (embit). This establishes **reference-vector consistency and independent cross-check (levels 1–2)** — not production or target conformance (levels 3–4), which no artifact yet reaches.

ENTROPY · 02		KEYS		DESCRIPTOR · 03		TRANSACTIONS · 04		RECOVERY · 05/07
dice / RNG → ENT	→	**ENT → master keys**	→	**keys → wallet**	→	**wallet → PSBTs**	→	**→ candidate spend**

`vs BIP-39 Trezor`
`vs RFC 5869`
`seed 5eb00bbd…`
		`vs secp256k1 73c5da0a`
		`→ fp 2fae9711`
		`fp 72a14ab8`
				`vs BIP-32 TV1`
				`vs bech32 BIP-173`
				`m/48'/0'/0'/2'`
						`embit cross-check`
						`reproduces desc`
						`ref corpus only`
								`embit partial sig`
								`final witness`
								`NOT yet proven`

The A/B/C fingerprints `2fae9711` / `72a14ab8` / `badaa8ff` (regenerated at 100 rolls) are shared across the QKEC-1 vectors, the Standards descriptor, and the normative **2-of-3** PSBT corpus; the retired 2-of-2 corpus survives only as a historical regression fixture. One computation, cross-reproduced by embit. A partial leg-A signature is *not* proof that a finalized recovery witness executes; that needs independent finalization and `testmempoolaccept` .

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03 SETTLED DECISIONS UNDER THE V1 ARCHITECTURE

Quick reference — design selected, implementation stop-ship

These are the fixed design decisions of the Core Architecture v1. `selected` means the design choice is made and normative; it does *not* mean built or validated. Implementation, hardware, and rehearsal evidence stays behind the gates below.

AREA	DECISION	STATUS	WHERE
Wallet policy	`2-of-3 · wsh(sortedmulti(2,A,B,C))`	`selected`	v1·05
Derivation	`m/48'/0'/0'/2'/{0,1}/*`	`selected · interop pending`	03

Seeds	24-word / 256-bit · no passphrase	selected	02
Entropy	QKEC-1 HKDF-SHA256 · 100-roll dice	ref vectors · review pending	02
Transport	microSD (binary) + QR (BBQr) · both	selected · QR perf = Gate A	04
PSBT	v0 profile · SIGHASH_ALL	selected · v2 deferred	04
Change proof	rebuild from descriptor D, mandatory	selected	04
A1 AEAD	ChaCha20-Poly1305 (RFC 8439)	selected · codec = Gate A	07
Signing	exact RFC-6979 + verify-before-release	selected · anti-exfil deferred	06
Cards	bearer · no PIN · account xprv · A2 on both	selected · feasibility = Gate B	06
Card channel	none in v1 (trusted card bus)	selected · SCP later hardening	06
Recovery	A1+B normal; A1+C / B+C recovery	selected · rehearsal = Gate E	05
Descriptor survival	D on both cards + cloaked kit copy	resolved	05
Core language	Rust + narrow libsecp FFI	hypothesis · Gate C	08
Network	mainnet only	selected	03

04 WHAT'S OPEN

The gates that must close before funding

The architecture is settled and two former stop-ships are resolved by the A1/A2/B/C model — but target-card feasibility, the A1 capture codec, the production core, storage/power behavior, and a full recovery rehearsal remain unresolved. The Core Architecture v1 sequences the evidence as five gates:

- ● **Resolved — Seed-A independence & descriptor survival.** A2 lives on both bearer cards, so A1 needs no key held only on the primary card; the complete checksummed descriptor D lives on both cards plus one redundant cloaked kit copy, so every valid pair can recover the wallet map.
- ○ **Gate A — A1 & QR on the fixed mechanics.** Freeze the ChaCha20-Poly1305 vectors and implement them twice; real disguised-page print/capture and human-fallback trials; zero unauthenticated plaintext under fuzzing; benchmark BBQr on the actual camera/optics/enclosure/320×240 display — no mechanical redesign.
- ○ **Gate B — card feasibility.** A real card capability matrix (secp256k1 sign, atomic transactions, timing, endurance, random source, power-cut); independent B/C account keys, same A2, role + policy/path enforcement; commodity-CCID-reader test; a published APDU spec and rescue vectors.
- ○ **Gate C — production core.** Rust target/toolchain decision closed and a reproducible build; differential tests vs Bitcoin Core / libsecp256k1 / an independent PSBT implementation; coverage-guided fuzzing; verified secret-memory plan; a hostile corpus far beyond seven fixtures; independent crypto/protocol review.
- ○ **Gate D — storage & power failure.** Power removal at every provisioning commit and SD write boundary; corrupt/full/missing SD and no-input-overwrite proof; card staged/live recovery; update interruption and rollback behavior.
- ○ **Gate E — complete recovery rehearsal.** A1+B, A1+C, and B+C all recover the same wallet and sign valid transactions; destroy/replace the terminal; a lost-factor full-wallet rotation into a fresh A′/B′/C′/A2′/D′ kit; recover with the open rescue tool and a commodity reader; `testmempoolaccept` on every reference recovery transaction.

> **PROVENANCE & HONEST LIMITS**
>
> Every hex value across these drafts was produced by pure-stdlib reference code and cross-checked against BIP-39 (Trezor), the canonical empty-passphrase seed, RFC 5869 (HKDF), secp256k1 (`73c5da0a`), the BIP-32 / BIP-173 test vectors, and embit. That establishes reference-model consistency and independent cross-check only. The Rust core, JavaCard applet, Linux I/O path, boot chain, and target hardware have **not** passed because a Python generator produced consistent values. This is engineering working material, not a security audit or a feasibility proof.

QuietKey QK2 Dossier · 2026–08–18 · under Core Architecture v1

Navigable index to the normative Core Architecture v1, the Decision Register (doc 00, the internal consistency matrix), and the eight QK2 working drafts. Open a document in a new tab from the cards above. Synthesised from primary-source reads of iancoleman/bip39, SeedSigner/seedsigner, and Coldcard/firmware, the QK2-04 & QK2-HW-P0/RV3 blueprints, and the documented loss history of this device class. Decisions are selected, not validated; production and hardware validation remain open.